

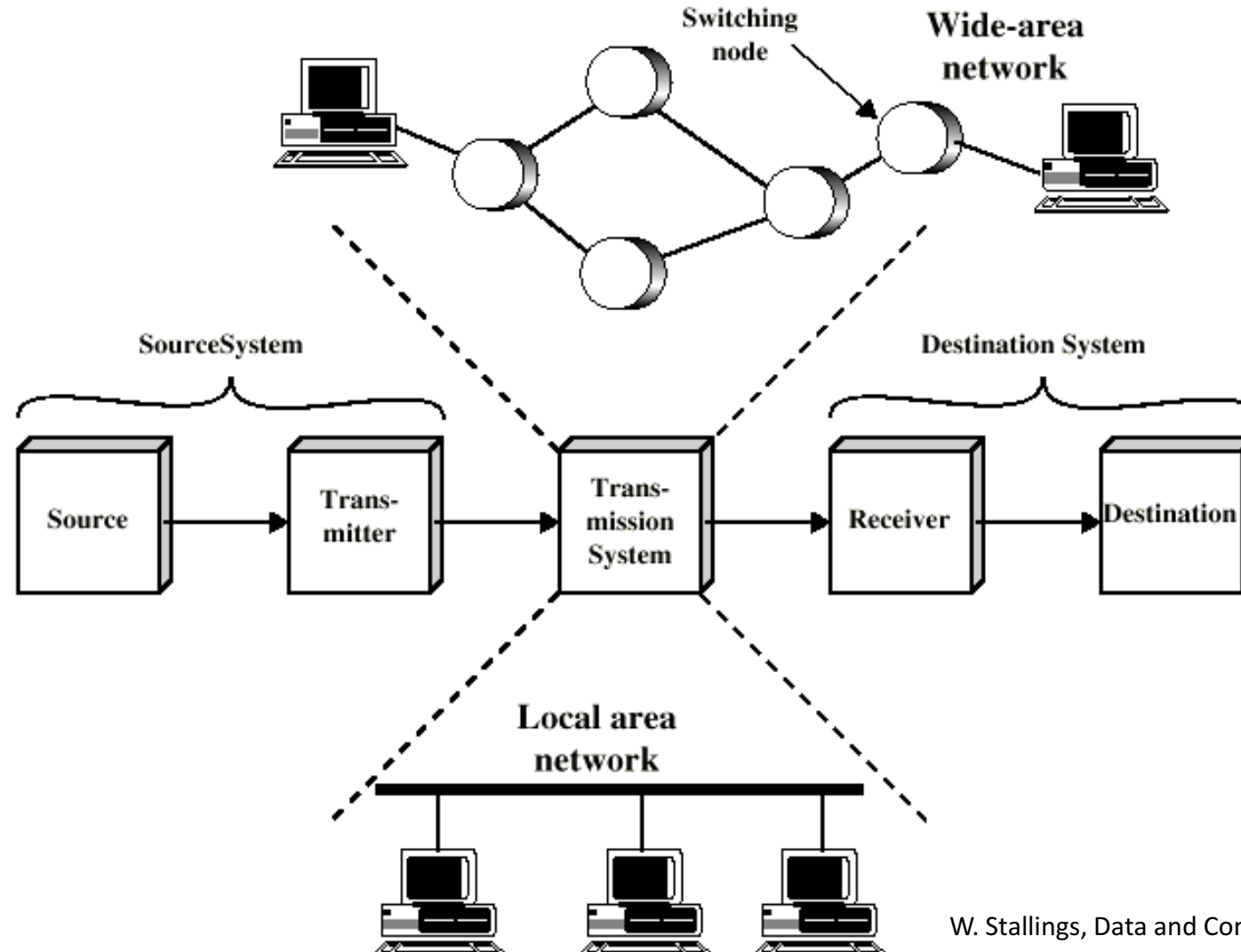
Basic Network Concepts and Web Technologies

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Outline

- Basic network concepts
- Basic WWW technologies
- Architectures of distributed applications

A Communication Model



Basic terms

- A **process** is a program running within a host.
 - within same host, two processes communicate using **interprocess communication** (defined by OS).
 - processes running in different hosts communicate with an application-layer protocol through a computer networks
- A **computer network** is a hardware and software data communication system that provides interconnection of computers and other devices.
- A **node (host)** is an addressable device attached to a computer network - computer, smart phone, sensor, network device, smart TV, smart kettle, etc.

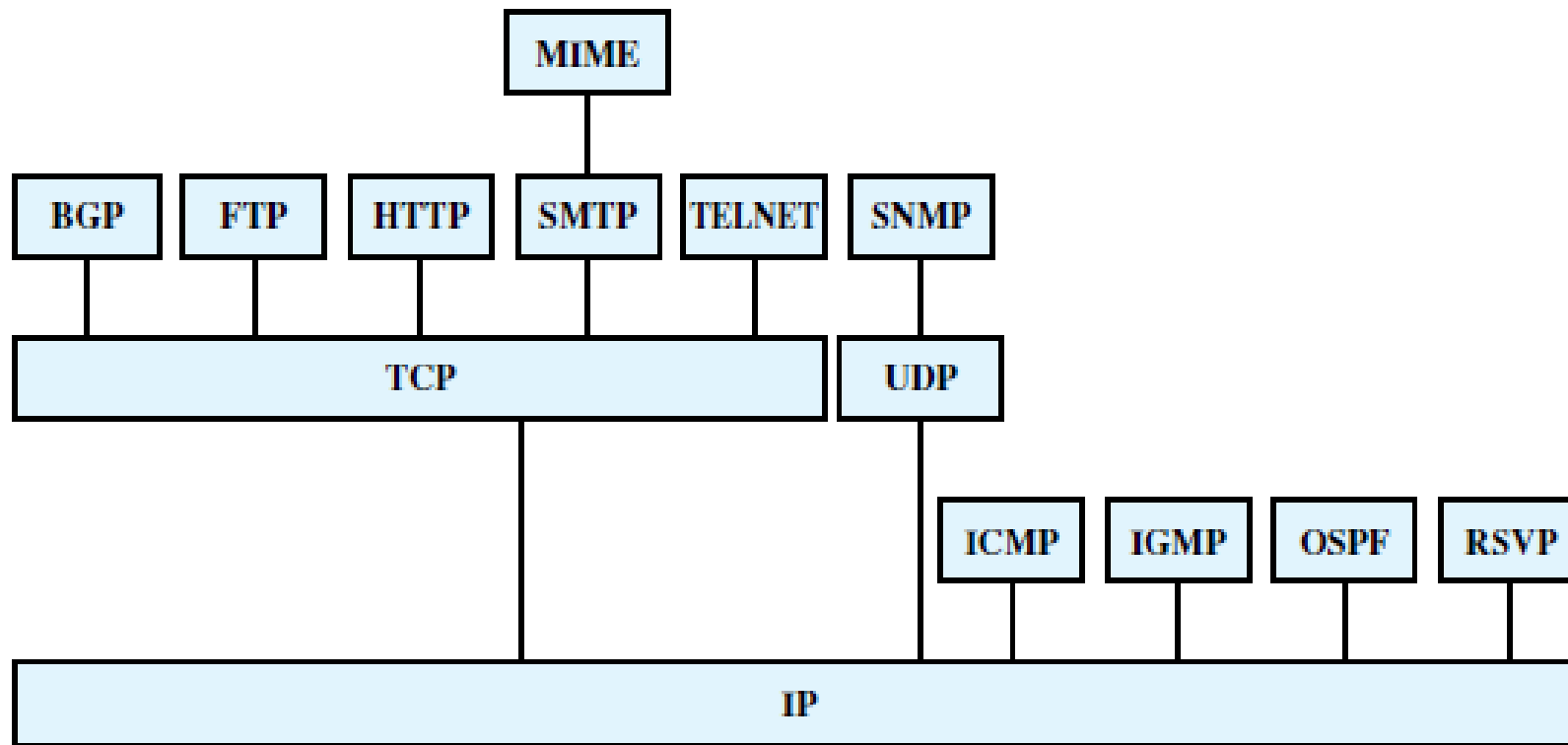
Basic terms 2

- For reducing a design complexity of the computer networks, they are organized in several **layers**, one above the other. The purpose of each layer is to offer **services** to the above layer, protecting it from the implementation details about the services. Layers from the same level can communicate to each other's using protocols.
- Generally, a **protocol** is a collection of rules, that determine how two or more participants in a given communication, can interact between them.
- A protocol defines types of messages exchanged (e.g., request & response messages); syntax of message types (what fields are in messages & how long is each field); semantics of the fields i.e., meaning of information in fields; and rules for when and how processes send & respond to messages
 - **Public-domain protocols** - defined in RFCs, allow for interoperability (e.g., HTTP)
 - **Proprietary protocols** – unknown in details for public (e.g., Skype)

Layered Protocol Architecture: OSI vs TCP/IP model

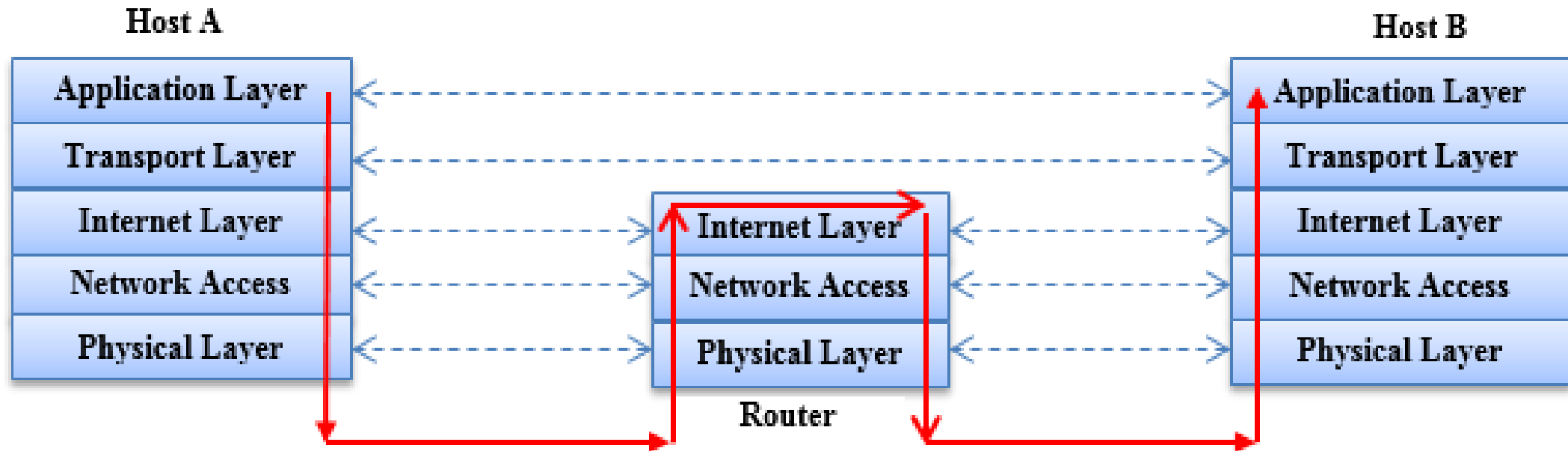
OSI	TCP/IP
Application	Application
Presentation	
Session	
Transport	Transport (host-to-host)
Network	Internet
Data Link	Network Access
Physical	Physical

Part of the upper three layers of the TCP/IP protocol suite



W. Stallings, Data and Computer Communications

Flow of Data in the Layered Protocol Architecture



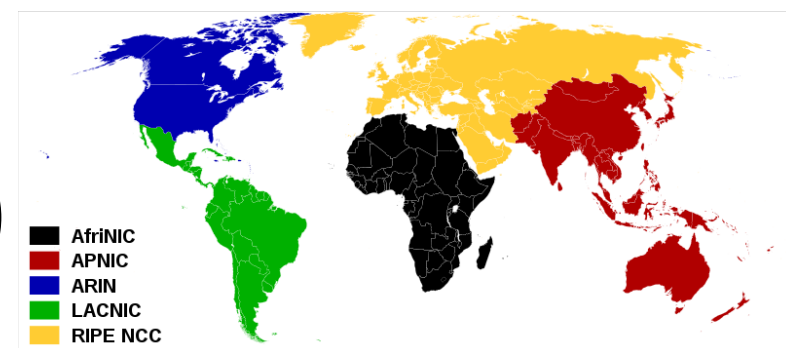
Protocol Encapsulation



Internet Protocol (IP)

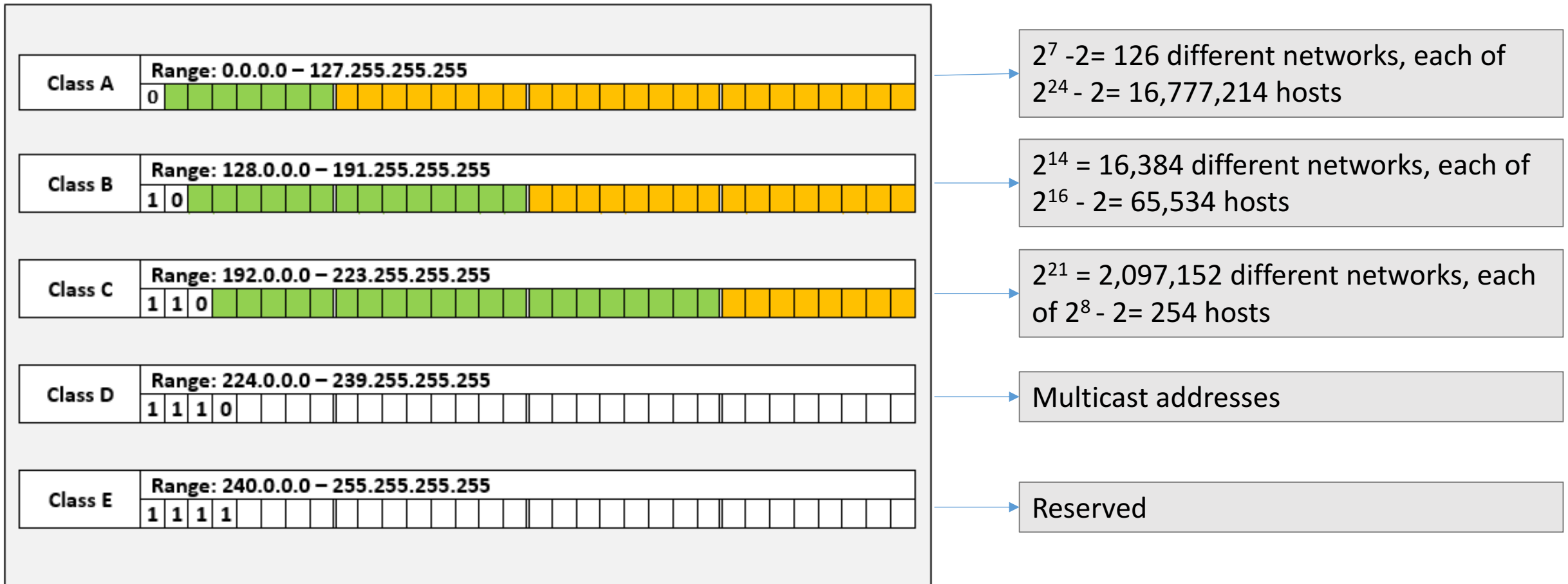
- The **Internet Protocol (IP)** is a connectionless internet layer protocol.
- Advantages of connectionless internet service
 - flexible
 - robust
 - not impose unnecessary overhead for connectionless transport protocols
- Unreliable
 - no guaranteed delivery
 - no guaranteed sequence of delivery
 - datagrams can take different routes
 - reliability is the responsibility of the next higher layer (e.g., TCP).
- IP addresses
- Two important versions coexist today: IPv4 (RFC 791) and IPv6 (RFC 1752, 2460, 2373, ...)

Internet Addresses (IP Addresses)

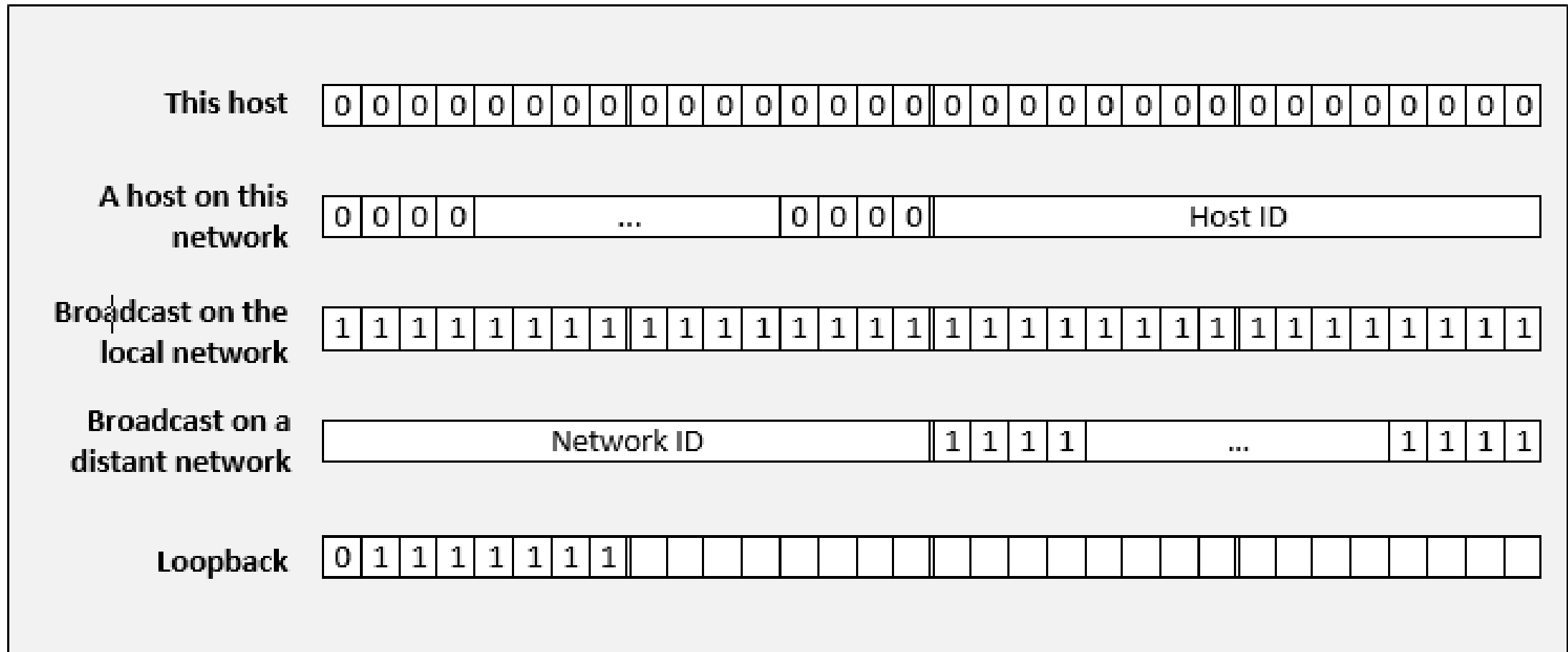


- An **Internet address (IP address)** can be IPv4 address or IPv6 address.
- **IPv4 address** of a node on the Internet is a unique 4-byte (32-bit) unsigned integer number.
 - *Dot decimal notation*: four unsigned integers, each ranging from 0 to 255, separated by periods.
 - **Example**: 11000000 11100100 00010001 0111001 is writing as 192.228.17.57
- **IPv6 address** of a node on the Internet is a unique 16-byte (128-bit) unsigned integer number.
 - Eight groups of 4 hexadecimal digits, separated by colons (:).
 - **Example**: 8000:0000:0000:0000:0123:4567:89AB:CDEF
- Internet addresses are assigned by Internet Corporation for Assigned Names and Numbers (ICANN) and its part Internet Assigned Numbers Authority (IANA) through Regional Internet Registries (AfriNIC, ARIN, APNIC, RIPE NCC, and LACNIC) and Local Internet Registries

Classful IPv4 addressing

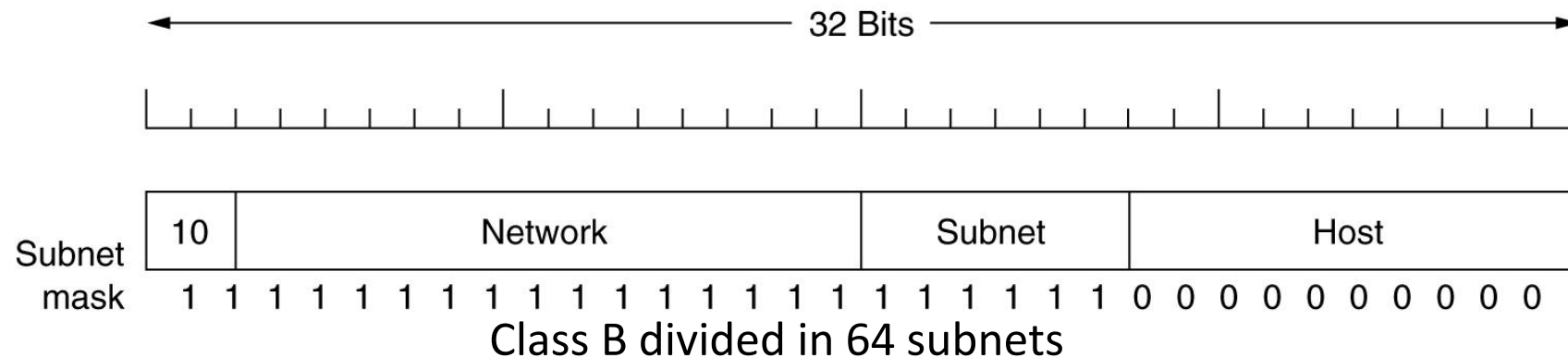


Some Special IPv4 Addresses



Classless Inter Domain Routing (CIDR)

- A method of IP address allocation and routing that allows more efficient use of IP addresses
 - CIDR notation: a.b.c.d/n (n = number of ones in the network prefix)
- To each LAN a subnet number is assigned
 - The host portion of the internet address is partitioned into a subnet number and a host number to accommodate this new level of addressing.
 - Local routers need to route packets to subnets
 - The address mask indicates which bits are from the subnet number, and which are from the host number.
- Variable Length Subnet Masking (VLSM)



Transport Layer:

Transmission Control Protocol (TCP)

- Transport layer protocols provide end-to-end communications between processes on different hosts
- **Transmission Control Protocol (TCP)** is a reliable connection-oriented stream-based transport protocol (RFC 793).
 - Allows sending data in a continuous stream
 - Guarantees delivering in proper order.
- Phases of TCP communication:
 - Establish a connection (open a TCP session) using three-way handshake
 - Transfer data over the connection
 - Release the connection
- Applications using TCP: file transfer, email, WWW

Transport Layer:

User Datagram Protocol (UDP)

- **User Datagram Protocol (UDP)**, is a unreliable connectionless transport protocol (RFC 768).
 - For pure message passing (datagram send/receive).
 - Neither guarantees delivery nor requires a connection.
 - Connectionless:
 - UDP datagrams are sent between two hosts with no previous setup.
 - The datagrams contain the destination address, may take different routes.
 - Lightweight and efficient. Low overhead compared to TCP
- Phases of UDP communication:
 - Sending: create a UDP socket; create a datagram with the message and specified destination (IP address & port); send the datagram over the UDP socket.
 - Receiving: receive a datagram from the UDP socket; get data and source from the datagram
- Applications using UDP: DNS, real time applications (IPTV, VoIP, videoconferencing, radio broadcast, online games, etc.).

Ports

- **Port** is a 16-bit transport layer address of a process that resides on a host.
 - 65,535 logical ports with integer numbers 1 - 65,535
 - TCP and UDP use source and destination port in their header, to identify processes on different nodes that exchange data
- A port can be allocated to a particular service:
 - A server listens the port for incoming requests
 - A client connects to the port and requests the service
 - The server replies via the port.
- Ports with numbers 1-1023 are reserved for well-known services.
 - A list of services and allocated ports is stored in
 - /etc/services (Linux)
 - C:\WINDOWS\system32\drivers\etc\services (Windows)

Part of the Assigned Ports

echo	7	tcp/udp	Echo back the input
discard	9	tcp/udp	Discard the input
daytime	13	tcp/udp	Output an ASCII string with the current time
ftp-data	20	tcp	Data port of ftp: transfer file
ftp	21	tcp	Command port of ftp: send ftp command
telnet	23	tcp	Interactive remote command-line sessions
smtp	25	tcp	“Simple Mail Transfer Protocol”: send email
time	37	tcp/udp	The number of seconds since Jan. 1, 1990
whois	43	tcp	Directory service for Internet administrators
finger	79	tcp	Information about a user or users
http	80	tcp	HyperText Transfer Protocol of WWW
pop3	110	tcp	Post Office Protocol for server-to-client mail
nntp	119	tcp	Network News Transfer Protocol
RMI	1019	tcp	Java RMI registry service

Host Names

- A **hostname** is a unique name of a computer on the Internet. It consists of a local name and a domain name.
- A given host may have multiple names



- One name can be mapped to multiple IP addresses



Domain Name System (DNS)

- **DNS** is a distributed service on the Internet that translates host names into IP addresses, and vice versa (RFC 1034/1035).
- **nslookup** - lookup IP-address by name (or vice versa)

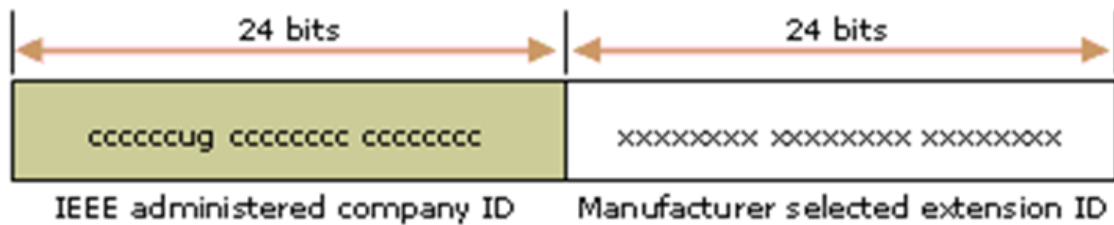
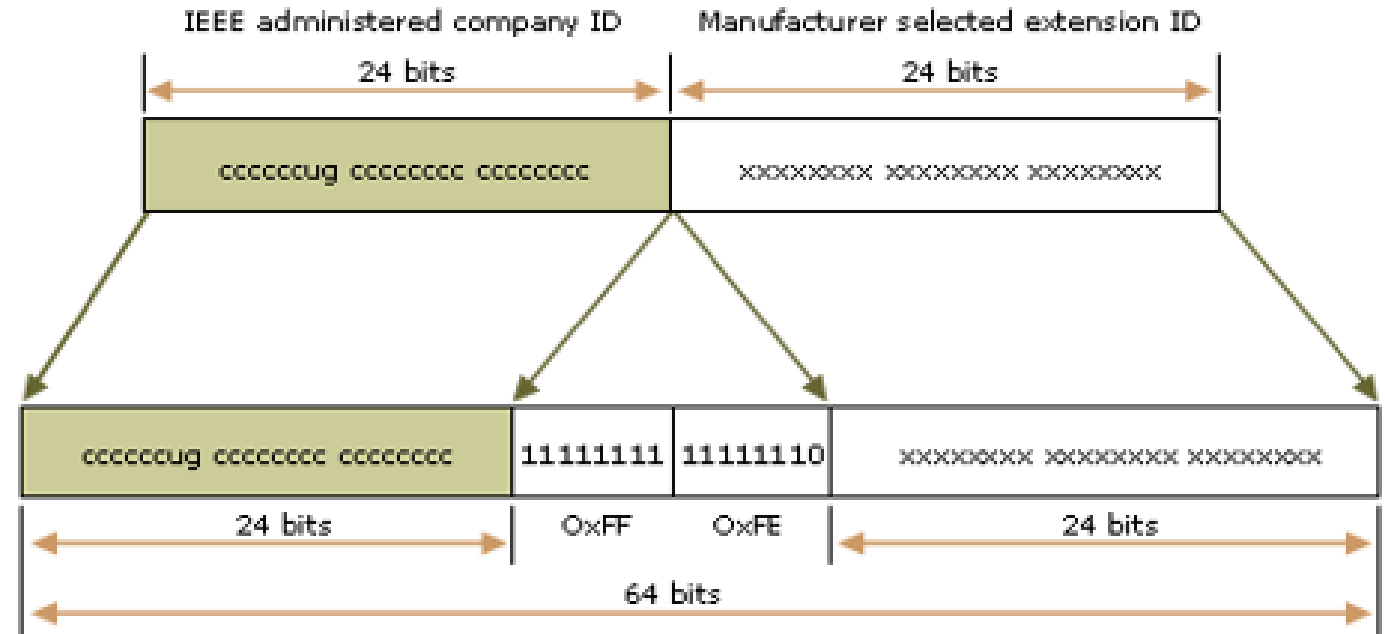
```
C:\>nslookup www.su.lt
Server:  ISAR.ugd.edu.mk
Address:  10.0.0.5

Non-authoritative answer:
Name:     web2.su.lt
Addresses:  2001:778:2200:200:f4d7:b6ff:fe46:dd03
           193.219.168.28
Aliases:  www.su.lt
```

MAC addresses

- **MAC (Media Access Control) address (Hardware or Physical) address**
 - *A unique identifier* assigned to a network interface controller (NIC) from manufacturers, for communications at the data link layer of a network segment.
 - Length of 48-bits, as 00-16-F0-67-89-ab or 64-bits
 - One node may have multiple NICs -> multiple MAC addresses.
- MAC address space is managed by the IEEE
 - vendors buy part of the MAC address space (<http://hwaddress.com/>)
 - **MAC-48** (obsolete), **EUI-48** and **EUI-64** (Extended Unique Identifier)
 - EUI-48 is used for most of the IEEE 802 network technologies (802.3, Bluetooth, 802.11, 802.5), FDDI, ATM, ...
 - EUI-64 is used for: IPv6, FireWire, wireless PAN, ...
 - Rule for converting MAC-48 and EUI-48 in EUI-64

MAC addresses



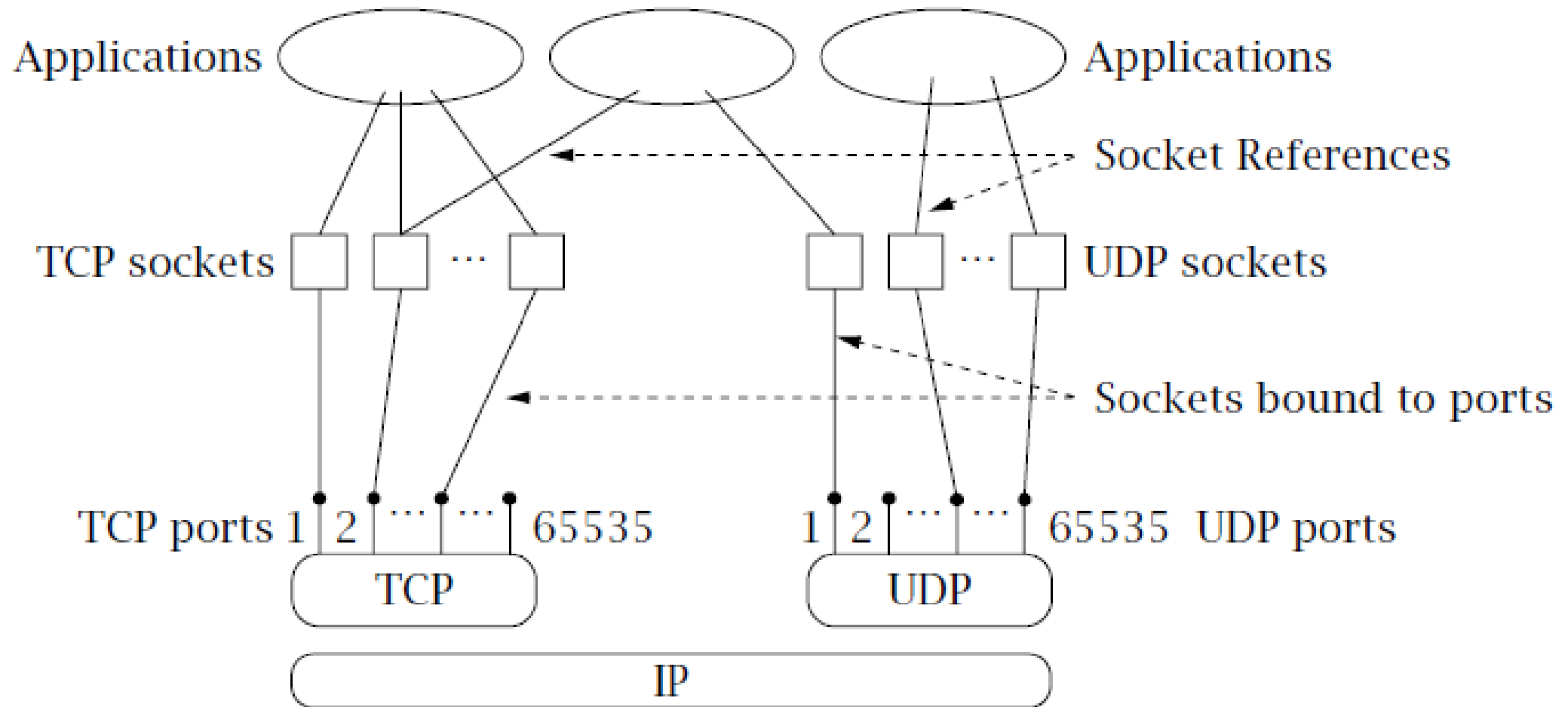
Client-Server Architecture

- Typical network application or service has two pieces: client and server
- Client:
 - initiates contact with the server (“speaks first”)
 - typically requests service from server,
 - Web: client implemented in browser; e-mail: in mail reader
- Server:
 - provides requested service to client
 - e.g., Web server sends requested Web page, mail server delivers e-mail

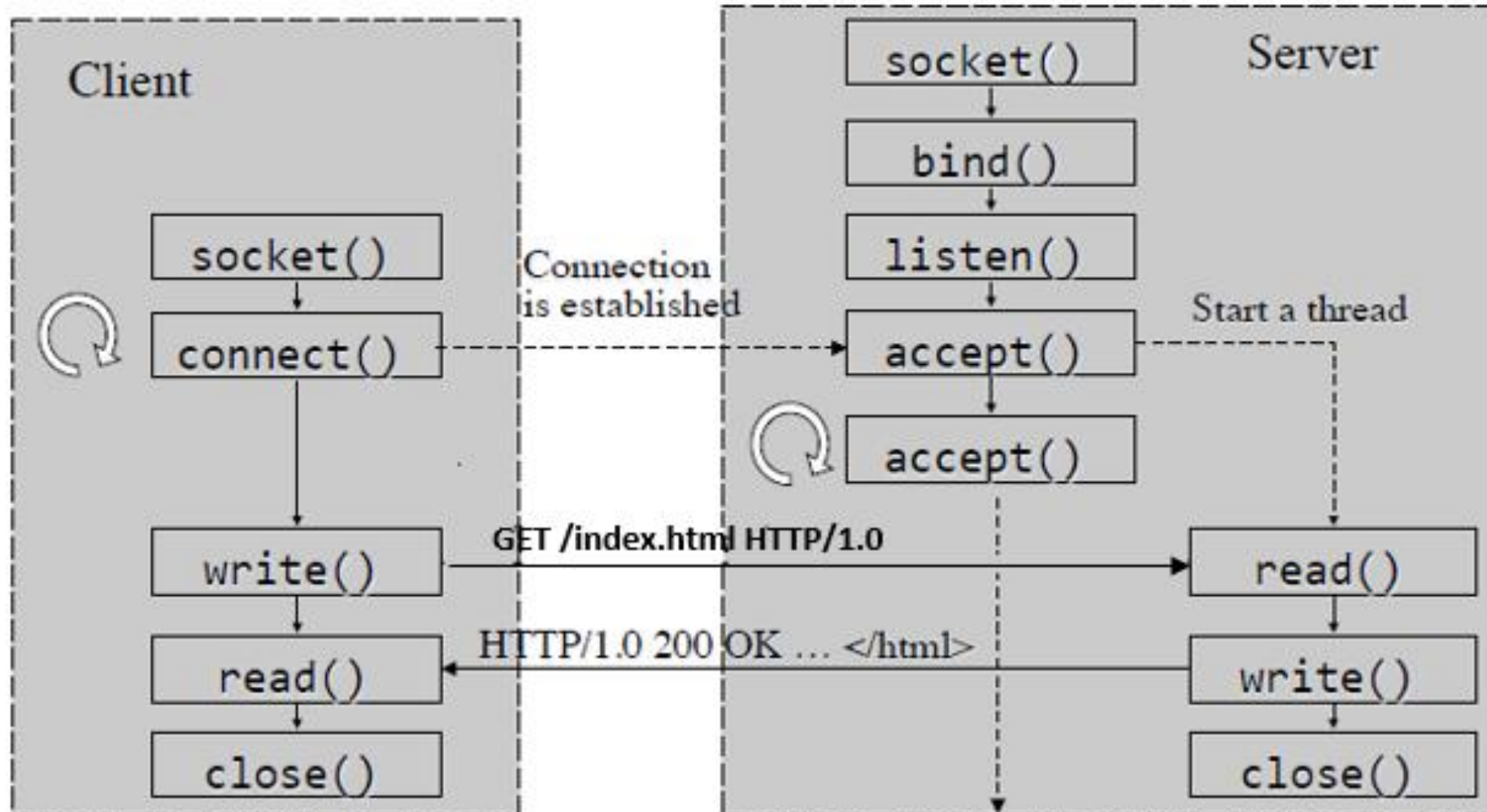
Sockets

- **Socket** is an abstraction through which a process may send and receive data
 - A socket address: combination of IP address and a port number
 - A socket type: distinguished by the transport protocol used for communication over the socket
 - TCP socket - stream-based, connection-oriented
 - UDP socket - datagram-based, connectionless
- The socket API in C, a.k.a. Berkeley sockets, was introduced in 1981 as the Unix BSD 4.2 generic API for inter-process communication
 - Initially was a part of the kernel (BSD Unix)
 - Today is a library

Sockets



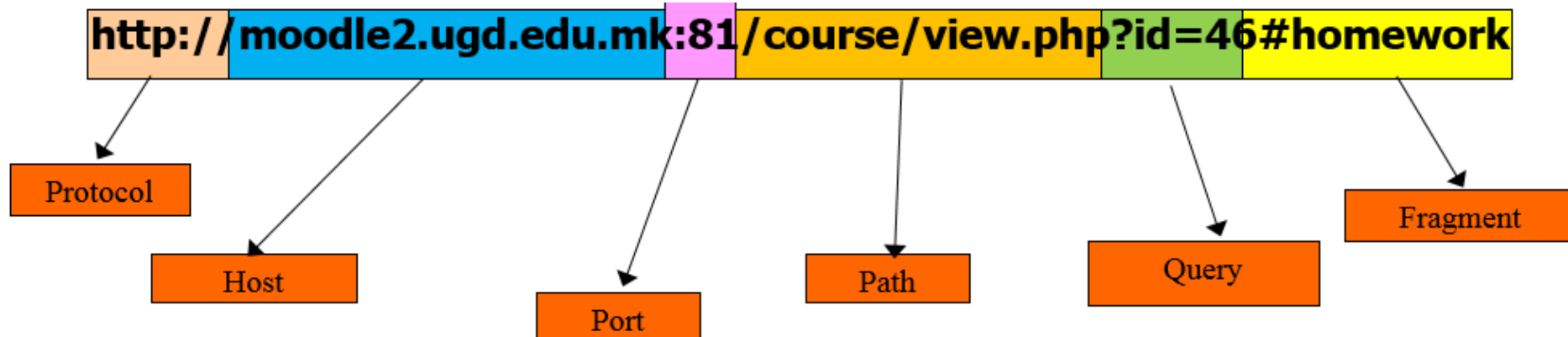
The Berkeley Socket API for the Client-Server Architecture



Some Basic Web Technologies

WWW, URIs, URLs

- **World-Wide Web (WWW, the Web)** is distributed client-server information system on the Internet
 - allows to locate and to access resources (files, services) on the Internet pointed on by URIs via servers by using Web protocols such as HTTP, FTP, Telnet, etc.
- **Uniform Resource Identifier (URI)** identifies a *resource* either by location or name. More often than not, most of us use URIs that defines a location (Uniform Resource Locator - URL) to a resource.



Some Web Protocols

- **Hyper Text Transfer Protocol (HTTP)**
 - A client-server request-response stateless protocol (RFC 2086 -v1.1, RFC 7540 – v2)
 - The most used HTTP requests are GET and POST
 - Server process: httpd; Default port: 80; Https port: 443 (HTTP on SSL/TLS)
- **File Transfer Protocol (FTP)**
 - A session-oriented out-of-band protocol for file transfer (RFC 959)
 - Ports: 20 (data), 21 (commands)
- **Simple Mail Transfer Protocol (SMTP)**
 - A server-to-server protocol for e-mail transfer (RFC 821)
 - Port: 25
- **Simple Object Access Protocol (SOAP)**
 - A protocol for exchanging structured information in the implementation of Web Services.
 - Relies on XML for message formats, and HTTP (SMTP) as a transport protocol for message transmission.

Some Markup Languages

- **HyperText Markup Language (HTML)**

- A Hypertext document format used on WWW.

- "Tags" are used to mark text elements:

< directive (case insensitive), zero or more parameters > text element </directive>

- Links to other documents:

foo

- HTML forms allow constructing a simple GUI embedded in an HTML document for a Web client.

- To input a user request and submit it to a CGI program or a servlet (JSP/JSF).

- **Extensible Markup Language (XML)**

- A language for exchange of a wide variety of data on the Web and elsewhere.

Dynamic Web Content: Server-Side Processing

- Provides dynamically generated contents on a HTTP request.
- **JavaScript** – a client/server-side script language
- **Server-Side Scripting**
 - Embedding program code in HTML documents, parsing and executing the code by the Web server; the result is included in the place of the code.
 - Examples:
 - **Active Server Pages (ASP.NET)** from Microsoft
Languages: C#, VBScript based on Visual Basic
Tags <% dynamic code %>
 - **Hypertext Preprocessor (PHP)** open source
A server-side, cross platform HTML-embedded scripting language
 - **Java Server Faces (JSF)** from Oracle
Language: JSF markup and Java

Dynamic Web Content: Server-Side Processing 2

- **Java Servlets**

- Provides extra functionality on the server side (extends web-server), process HTTP requests from HTML forms and applets, and generate content
 - For example, provides access to corporate databases and information services in the third tier of a 3-tier application
- A servlet lives in the server's JVM (application server) much like an applet lives in the client's JVM (in Web browser).
- Once created, a servlet is alive as long as the server (JVM) is alive
- The servlet can keep state between requests
 - New state and response is a function of old state and request
- A servlet can be multithreaded – Scalability
- A servlet may use EJBs – Extendibility
- The Java Servlet API is available as a part of Java Platform, Enterprise Edition (javax.servlet), and together with JSFs are supported on many web servers

- **Enterprise JavaBeans**

Architecture of Distributed Applications

Distributed Applications

- Motivation:
 - Data, computers and resources, users (clients) are geographically distributed;
 - Improve performance or/and scalability or/and robustness of applications by means of distributed execution.
- Distributed applications on a network of computers (LAN, WAN, the Internet):
 - Print servers, distributed file systems (DFS), DNS, rlogin;
 - WWW: web servers and browsers, ftp and mail servers, ftp and mail clients, instance messaging, on-line games, content delivery networks, streaming media applications, web-services, etc.;
 - Financial and commercial applications: E-commerce, banking (OLTP);
 - Remote control and monitoring of networked devices;
 - Scientific and engineering computing;
 - Cloud computing environments;
 - Content delivery (or distribution) networks (CDN)

Basic Architectures of Distributed Applications

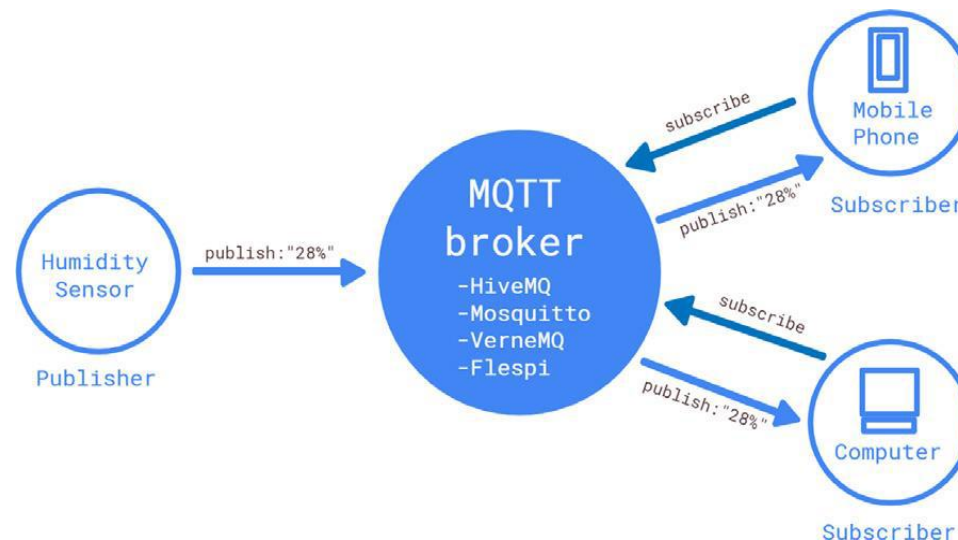
- **Two-tier architecture** (a.k.a. **client-server** architecture)
- **Publish/subscribe model**
- **Three-tier architecture**
 - Clients (with UI, GUI) – in the 1st tier
 - Business logic – in 2nd tier
 - System services (databases) – in the 3rd tier
- **Peer-to-peer (P2P) architecture**
 - Formed of peers– processes running on networked nodes
 - All peers are equal, being both clients and servers
- **Service-Oriented Architecture (SOA)**
 - Builds on web-services with well-defined interfaces, which can be described, deployed, discovered, bound, composed, invoked.

Problems of 2-tier Client-Server Architecture

- Portability
 - No control over the client operating system and hardware.
 - Challenging to upload anything to the client if it does not accept.
- Efficiency
 - A “fat” client may require too much resources on a client machine
- Also slow to download (applets)
 - Direct SQL access can generate lots of network requests
- Security – the most important
 - DBAs do not accept the risks of putting the database on the Internet
 - Internet security should be at the service level, not at the data level

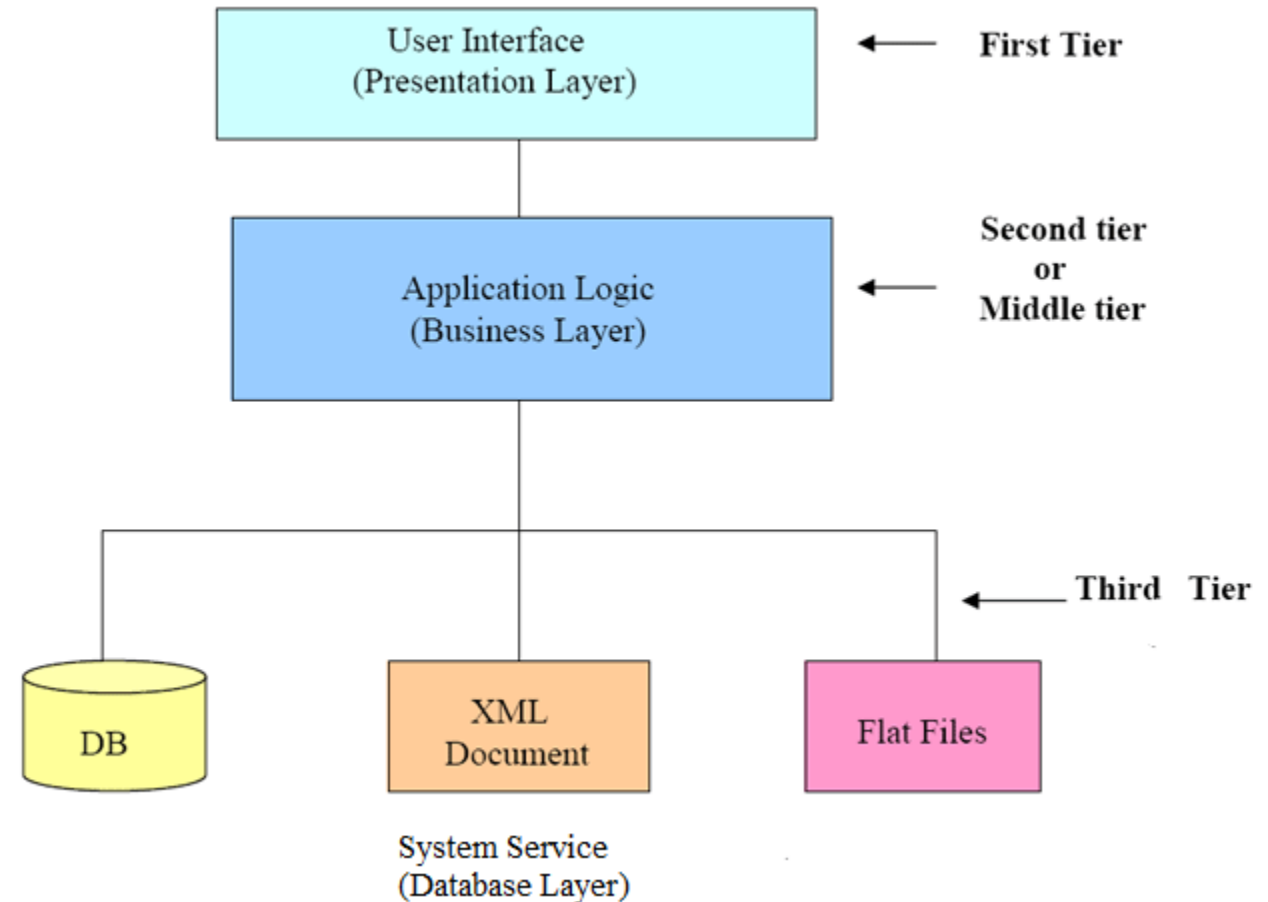
Publish/Subscribe Model

- Basically, this is a client/server model that allows the client to communicate with an endpoint.
 - The clients that send messages are called *publishers*.
 - Clients that receive the messages are called *subscribers*.
 - A central point that plays the role of a server and is called a *broker*.
- • Example, Message Queuing Telemetry Transport (MQTT)



3-Tier Architecture

- User-Interface Tier
 - The layer of user interaction.
 - A “thin” client of the business logic servers
- Business Logic Tier
 - It is made up of business objects: inventory control, budget, transaction monitors, ORBs, authentication, etc.
- System services Tier
 - Objects that encapsulate database routines and interact with DBMS.



Advantages of the 3-Tier Architecture

- Improved performance
 - Use faster protocols than HTTP or ODBC
 - Download the GUI (thin client), but leave the rest of the logic on the server or in the middle-tier
- Manage security
 - The middle-tier can control user authentication, access to resources in the third tier, applet security rules
- Manage user application context
 - The server can remember user data
 - The user can access his context from any Web client

Peer-to-Peer (P2P) Architecture

- A P2P application runs on an overlay network
 - All peers are equal in terms of responsibility, capabilities and functionality: typically execute the same set of algorithms, participate in distributed algorithms
- An **overlay network** is a “virtual” network of nodes created on top of an existing network, e.g. the Internet.
 - Each node has an ID, knows neighbors, does not know the global topology, communicates as a source and a destination, and also **serves as a router** in sending data.
- Structured overlay (P2P) networks
 - Common example is by using **Distributed Hash-Table (DHT)**
 - E.g. Chord, Pastry, Tapestry
- Unstructured overlay networks
 - E.g. Gnutella

Basic Communication Mechanisms

- Message passing over sockets (TCP or UDP)
 - Application specific request/response protocols
- Remote Procedure Calls (RPC) and rendezvous
 - RPC spawns a new process (thread) to handle a request
 - Rendezvous – request is accepted (selected) and processed by an existing running server process
- Remote Method Invocation (RMI)
 - The object-oriented analog of RPC in a distributed object-oriented environment
 - Distributed object architecture

SOA: Service-Oriented Architecture

- Applications are built of services
 - Services are built of components
 - Components are bound to each other via client/server interfaces
- A client interface of a (client) components is bound to a server interface of another (server) component
- Services are loosely-coupled
 - Expose interfaces (port types);
 - Can be described, discovered, bound, and invoked;
 - Service invocation: request-response interaction

Some Existing Approaches

- CORBA - outdated
 - Common Object Request Broker Architecture from OMG
 - Heterogeneous
 - Many implementations exist
- .NET (DCOM)
 - Distributed Component Object Model from Microsoft
 - Homogeneous (“MS-only”)
- Java RMI
 - Homogeneous
 - Enterprise JavaBeans – A component architecture for building integrated enterprise services based on RMI/IIOP
- Web services
 - SOAP (Simple Object Access Protocol) – a minimal set of conventions and standards for invoking code using XML over HTTP